

DIGITAL FORENSICS

INVESTIGATION

Organization: Apex Integrated Technologies Ltd.

Case ID: AIT-DF-2026-009

Investigation Title: Digital Forensic Investigation-Deleted File Recovery

Investigator Name: Chikere James

Date : Friday 25th September 2026

EXECUTIVE SUMMARY

This investigation examined a forensic image (Evidence.E01) supplied by Apex Integrated Technologies Ltd. in relation to the reported disappearance of six folders: SSH, My pic, Our pic, ADDS Project, Corel, and Corel Draw Work .A working copy of the image was created and its integrity verified. Examination with FTK Imager and Autopsy identified deleted directory entries for SSH, Corel, and Corel Draw Work. The contents of these three folders were recoverable. The remaining three folders could not be located or recovered.

The file system was identified as NTFS. No evidence was found that attributes the deletion to a specific individual or that establishes intent. A hash discrepancy between local calculation and Autopsy, and a size discrepancy in the evidence file, are noted as limitations.

The examination was limited to the supplied image and the tools specified in the assignment.

CASE BACKGROUND

Apex Integrated Technologies Ltd. discovered during a routine review that several folders previously present on a workstation were no longer visible. Management requested a forensic examination of the supplied disk image to determine what data had been present, what had been deleted, and what could still be recovered.

INVESTIGATION OBJECTIVES

The Primary objective is to examine the forensic image that was given to us by the organization and investigate the disappearance of the following folders:

- SSH
- My Pic
- Our Pic
- ADDS Project
- Corel
- Corel Draw Work

And also determine whether these folders are Present ,Deleted ,Partially deleted, Recoverable and Unrecoverable. Determine the timeframe in which the deletion occurred.

SCOPE

This examination was limited to the forensic image supplied by the organization (Evidence.E01). The investigation focused on determining the status of the six named target folders (SSH, My pic, Our pic, ADDS Project, Corel, and Corel Draw Work), whether they were present, deleted, or recoverable, and what contents could be recovered.

The following were examined:

- Working copy creation and integrity verification
- Filesystem structure and deleted directory entries
- Recoverable contents of the identified folders
- Cross-tool comparison of findings

The following were outside the scope of this examination:

- Attribution of the deletion to a specific user
- Determination of intent
- Full recovery or analysis of every file on the image
- Network or memory analysis
- Live system examination

EVIDENCE IDENTIFICATION

EVIDENCE INFORMATI ON	RESULT	SCREENSH OT REFERENC E
Evidence filename	Evidence.E01	Screenshot figure 2(Appendic es - section 4)
Evidence size	Initially documented as 10 MB. Autopsy reported approximately 52.4 GB (52,428,880,000 bytes). Discrepancy noted.	Screenshot figure 4 (Appendice s - section 4)
Image format	EnCase Image File Format	
Partition	None identified	
File system	NTFS	Screenshot figure 2(Appendic es - section 4)

User accounts	None Identified	
Acquisition date/time	2026-09-24T15:34:00+01:00	Screenshot figure 4(Appendices - section4)
SHA-256	6216149fddf6eb7074349afaef226f02c3b246ee6c3c3ae91c5ab7adeb8c29e0	Screenshot figure 1(Appendices - section 4)
MD5	9e2b4bb9adf2b77b0810f8f6bb115a3f	Screenshot figure 1(Appendices - section 4)

Hash Verification Observation

During the examination, different hash values were recorded for the forensic evidence. The SHA-256 value recorded during the initial verification was:

SHA-256:
6216149fddf6eb7074349afaef226f02c3b246ee6c3c3ae91c5ab7adeb8c29e0

The MD5 value recorded during the initial verification was:

MD5: 9e2b4bb9adf2b77b0810f8f6bb115a3f

However, Autopsy reported the following MD5 value for the evidence:

Autopsy MD5: ae39e9a30ab5a65809eb93289938e149

This difference was recorded as an evidence-integrity observation. The discrepancy alone does not establish that the forensic evidence was altered or tampered with, because the examination notes do not establish that both values were calculated from exactly the same underlying data object and scope.

The difference is therefore documented for transparency and further verification. The SHA-256 value was retained as the primary integrity reference because SHA-256 provides a stronger collision-resistance property than MD5. The differing MD5 values should be investigated by confirming the exact object and data scope hashed by each tool.

TOOLS USED

- DD
- FTK IMAGER

- **R DRIVE**
- **AUTOPSY**
- **VIRUSTOTAL**

METHODOLOGY

1. A working copy of the original forensic image was created using the dd utility to preserve the original evidence.
2. SHA-256 and MD5 hashes of the original image and the working copy were calculated and compared to verify integrity.
3. The working copy was examined in FTK Imager to identify partitions, filesystem structure, and the six target folders.
4. The same image was examined in Autopsy to analysis deleted data, recover folder contents, and record metadata.
5. An attempt was made to open the image in R-Drive Image; the tool was unable to load the image.
6. Findings from the tools were compared and documented.
7. Limitations, including hash and size discrepancies, were recorded.

FTK-IMAGER INVESTIGATION/FINGINGS

FTK Imager was used to examine the forensic image and identify the relevant partitions, filesystem structure, directories, files, and deleted data. The examination identified three of the six target folders: **SSH, Corel, and Corel Draw Work**. The remaining three target folders—**My pic, Our pic, and ADDS Project**—were not identified during the examination.

The identified folders contained recoverable files and directory structures, indicating that the folders and their contents had previously existed on the examined filesystem and that deleted data remained recoverable from the forensic image.

The available evidence does not establish which user or individual performed the deletion. Therefore, the findings are reported as evidence of deleted and recoverable data rather than as proof that a particular employee deleted the folders.

R-DRIVE FINDINGS

R-Drive Image was used as part of the forensic examination. The supplied forensic image could not be successfully identified or loaded by R-Drive Image during the examination.

As a result, no additional forensic findings or recovered files could be obtained from R-Drive Image. The reason for the incompatibility could not be established from the available evidence.

This limitation was documented rather than treated as evidence that the forensic image was damaged or invalid. Findings obtained from FTK Imager, DD, and Autopsy were therefore used for the remaining analysis and cross-tool validation.

DD FINDINGS

DD is an old Linux based tool that is used to acquire a forensic image/copy of a drive or a partition. Command used “**dd if=Evidence.E01 of=working-copy.E01 bs=4M status=progress**” this command was used to acquire a working copy of the forensic image that presented by the organization, this was done to preserve the original forensic image. An integrity verification was also done to confirm that both the forensic image and working copy were not altered or changed in anyway using the following these hashing algorithms; SHA-256 and MD5.

AUTOPSY FINDINGS

Autopsy was used to examine the forensic image, filesystem structure, deleted data, file metadata, and recoverable files.

The examined filesystem was identified as **NTFS**. The presence of an NTFS filesystem describes the filesystem structure contained within the forensic image; however, the filesystem type alone does not establish which individual created, modified, or deleted the files.

Autopsy identified three of the six target folders: **SSH, Corel, and Corel Draw Work**. These folders contained deleted data that remained available for recovery. The other three target folders—**My pic, Our pic, and ADDS Project**—were not identified as recoverable during the examination.

Autopsy also provided metadata associated with the forensic image, including the image format, sector size, acquisition information, and hash values. These metadata values were recorded as part of the examination.

- **Filename:** /img_Evidence.E01
- **Type:** E01

- **Size:** 52428880000
- **MD5 HASH:** ae39e9a30ab5a65809eb93289938e149
- **SHA1 HASH:** 4462d8bcc58431744669af554ce9046fee9b16bd
- **SHA-256 HASH:** Has not been calculated
- **Sector Size:** 512
- **Time Zone:** Africa/Lagos
- **Case Number:** aa
- **Acquired Date:** Thu Sep 24 12:17:52 2026
- **System Date:** Thu Sep 24 12:17:52 2026
- **Acquiry Operating System:** Win 201x
- **Acquiry Software Version:** ADI8.3.0.27
- **Device ID:**068fcf15-2e5c-44c0-8055-df01f9ac2f68

This is the metadata that was acquired from autopsy during the investigation, but during the investigation it has been shown that the hash of MD5 that was calculated using autopsy is different from the computed hash that was made using Kali Linux. Lastly, here is the timeline of how the investigation was carried out.

DELETED DATA RECOVERY

During the investigation using the tools FTK imager and Autopsy it was proved that only three out of the six folders were recoverable while the other three were unrecoverable. The three folders that were recoverable are:

- SSH
- Corel
- Corel Draw work

The unrecoverable folders are:

- ADDS Project
- My pic
- Our Pic

And it was not just the folders but also the content/files of the three folders were recovered as well.

TARGET FOLDER ANALYSIS

A. SSH

- Status: Deleted directory entry identified; folder and contents recoverable.
- Contents: Multiple files recovered, including documents related to Cron automation, firewall configuration, IOC investigation, network configuration, SSH

attack simulation, detection scripts, log investigation, SSH service configuration, UFW rules, and validation/testing materials. Most recovered files were in PDF format.

- Recovery: Folder and contained files successfully recovered using FTK Imager and Autopsy.

- Attribution: No evidence established the identity of the person who deleted the folder.

B. My pic

- Status: Not located as an allocated or recoverable directory entry.

- Recovery: Unrecoverable within the scope of this examination.

C. Our pic

- Status: Not located as an allocated or recoverable directory entry.

- Recovery: Unrecoverable within the scope of this examination.

D. ADDS Project

- Status: Not located as an allocated or recoverable directory entry.

- Recovery: Unrecoverable within the scope of this examination.

E. Corel

- Status: Deleted directory entry identified; folder and contents recoverable.

- Contents: Contained a subfolder named “Corel content” with multiple subdirectories including Adjustments presets, CorelDRAW, Export presets, Fills, Find and replace, Fonts, Image lists, Images, Painter, Palettes, Photo frames, Symbols, Templates, and Writing Tools. Some subdirectories were empty.

- Recovery: Folder structure and available contents successfully recovered.

F. Corel Draw Work

- Status: Deleted directory entry identified; folder and contents recoverable.

- Contents: Approximately 90 items recovered. File types included JPG, PNG, and CDR (CorelDRAW) files.

- Recovery: Folder and majority of contents successfully recovered.

CROSS TOOL VALIDATION

The findings from the various tools will be compared here:

FINDINGS	DD&R-DRIVE	FTK IMAGER	AUTOPSY	FINAL INTERPRETATION
Working copy creation and integrity.	Working copy created with dd; hashes matched.	Not applicable.	Not applicable.	Evidence integrity verified at time of working copy creation.
Detection of target folders.	R-Drive was unable to detect/open the forensic image that was given.	Identified folders SSH, Corel, Corel draw work.	Identified folders SSH, Corel, Corel draw work.	Three folders confirmed deleted and recoverable.
Recovery of folder contents.	Not performed.	Contents of the three folders recoverable.	All contents/files were recovered during the process.	Recovery successful for SSH, Corel and Corel draw work.
File system Identification.	Not performed.	NTFS	NTFS	File system confirmed as NTFS.
Hash consistency.	Local MD5/SHA-256 calculated.	Not calculated in this examination.	Reported different MD5 value.	Hash discrepancy noted as a limitation.

OVERALL FINDINGS

The examination established the following facts based on the available evidence:

1. A working copy of the forensic image was created using dd and its integrity was verified through matching SHA-256 and MD5 hashes.
2. R-Drive Image was unable to open or process the supplied E01 image; no findings were obtained from this tool.
3. FTK Imager and Autopsy both identified deleted directory entries for the folders SSH, Corel, and Corel Draw Work.
4. The contents of these three folders were recoverable.
5. The folders My pic, Our pic, and ADDS Project were not located as allocated or recoverable items.
6. The file system of the image is NTFS.
7. A discrepancy was observed between the MD5 hash calculated locally and the MD5 value reported by Autopsy. This remains unresolved.

LIMITATIONS

The following limitations apply to this investigation:

1. R-Drive Image could not open the supplied forensic image; therefore no findings were obtained from that tool.
2. A discrepancy exists between the MD5 hash calculated on the working copy and the MD5 value reported by Autopsy. The reason for this difference was not determined.
3. Three of the six target folders (My pic, Our pic, and ADDS Project) could not be located or recovered.
4. No evidence was identified that establishes the identity of any individual who performed the deletion.
5. No evidence was identified that establishes whether the deletion was intentional or accidental.
6. Detailed MACB timestamps for individual recovered files were not fully documented in this report.
7. The reported size of the evidence file differed between initial documentation (10 MB) and Autopsy metadata (approximately 52.4 GB). This discrepancy was not resolved.

CONCLUSION

The digital forensic examination identified three of the six target folders: **SSH, Corel, and Corel Draw Work**. These folders contained deleted data that remained recoverable from the forensic image. The contents recovered from these folders were documented as part of the examination.

The remaining three target folders—**My pic, Our pic, and ADDS Project**—were not identified as recoverable during the examination.

DD was used to create and verify a working copy of the evidence, while FTK Imager and Autopsy provided significant findings concerning the filesystem structure, deleted folders, and recoverable files. R-Drive Image was unable to successfully load the supplied forensic image, and this limitation was documented.

The examination supports the conclusion that the three identified folders and their associated data had previously existed on the examined filesystem and that deleted remnants remained recoverable. However, the available evidence does not establish which individual deleted the folders or provide sufficient evidence to assign responsibility for the deletion to a specific user.

The difference between the MD5 value calculated during the examination and the MD5 value reported by Autopsy was also documented. The available information is insufficient to conclude that the evidence was altered; further verification of the exact objects hashed by each tool would be required.

Overall, the findings demonstrate that deleted data from three of the six target folders remained recoverable, while the other three target folders were not recovered during the examination.

RECOMMEDATION

1. Maintain the original forensic evidence in a protected, read-only state and conduct further examination only on verified working copies.
2. Record SHA-256 hashes for the original evidence, working copies, and significant exported/recovered files to support evidence integrity.
3. Where multiple forensic tools produce different hash values, verify that each tool is hashing the same underlying object and the same scope of data before drawing conclusions about evidence integrity.
4. Preserve the recovered files in a designated evidence directory and avoid modifying the recovered files during subsequent analysis.
5. Maintain a complete chain-of-custody record documenting evidence collection, transfer, storage, examination, and export activities.
6. Maintain a detailed investigation log showing the date, time, examiner, tool used, action performed, and resulting observation for each significant forensic activity.
7. For future investigations, document filesystem timestamps and other relevant forensic artifacts separately from examiner activity timestamps so that investigation activities are not confused with the time of the underlying event.
8. Where possible, use multiple forensic tools to independently validate significant findings and clearly document any differences between tool outputs.
9. Preserve screenshots and supporting forensic records for significant findings so that another examiner can reproduce and independently evaluate the examination.

INVESTIGATION LOG

DATE/TIME	INVESTIGATOR	TOOL	ACTION	RESULT
24 Sep 2026, 15:34	Chikere James	–	Received forensic image Evidence.E01.	Image Received.

24 Sep 2026, 18:12	Chikere James	DD	Created working copy.	Working copy created; hashes matched.
25 Sep 2026, 13:40	Chikere James	FTK Imager	Examined image for target folders.	Identified SSH, Corel, Corel Draw Work as deleted/recoverable.
25 Sep 2026, 14:55	Chikere James	Autopsy	Examined image, recovered folders.	Confirmed NTFS; recovered contents of three folders.
25 Sep 2026, 16:35	Chikere James	R-Drive	Attempted to open image.	Image could not be loaded.
26 Sep 2026, 12:23	Chikere James	-	Documented findings and limitations.	Report prepared.

APPENDICES

1. CHAIN OF CUSTODY:

FIELD	INFORMATION
Case ID	AIT-DF-2026-009
Evidence ID	Evidence.E01
Evidence description	Forensic disk image (E01 format)
Received from	Apex Integrated Technologies Ltd.
Received by	Chikere James
Date/Time received	24 September 2026
Original Hash (SHA-256)	6216149fddf6eb7074349afaef226f02c3b246ee6c3c3ae91c5ab7adeb8c29e0
Storage Location	Working copy used for examination
Current Custodian	Chikere James

2. EVIDENCE PROCESSING TABLES:

EVIDENCE	ORIGINAL PATH	STATUS	CREATED	MODIFIED	ACCESS ED	RELEVANT/DELETED TIME	HASH	TOOL
SSH	/img_Evidence.E01/SSH	DELETED	2026-09-24,	2026-09-24,	2026-09-24,	Not Identified	Not Calculated	Autopsy

			12:39:46 WAT	12:45:41 WAT	12:45:41 WAT			
Corel	/img_Evidence.E01/Corel	DELETE D	2026-09-24, 12:40:08 WAT	2026-09-24, 12:40:08 WAT	2026-09-24, 12:40:08 WAT	Not Identified	Not Calculated	Autopsy
Corel draw work	/img_Evidence.E01/Corel draw work	DELETE D	2026-09-24, 12:42:36 WAT	2026-09-24, 12:42:36 WAT	2026-09-24, 12:42:36 WAT	Not Identified	Not Calculated	Autopsy

3. **TIMELINE:**

DATE	TIME	EVENT	EVIDENCE/PATH	SOURCE
2026-09-24	12:39:46	Folder created	SSH	Autopsy
2026-09-24	12:40:08	Folder created/modified/accessed	Corel	Autopsy
2026-09-24	12:42:36	Folder created/modified/accessed	Corel draw work	Autopsy
2026-09-24	12:45:41	Folder modified/accessed	SSH	Autopsy
2026-09-24	-	Deleted directory entries observed	SSH, Corel, Corel draw work	Autopsy/FTK Imager

Interpretation of the Timeline:

The recoverable timestamps for the three identified folders (SSH, Corel, and Corel Draw Work) all fall on 24 September 2026. This is also the date associated with the acquisition of the forensic image.

The evidence supports that these folders existed on the filesystem and that deleted directory entries remained recoverable at the time of examination. However, the available timestamps do not provide a precise moment of deletion that can be separated from the creation/modification activity or from the acquisition process itself.

Therefore, the most defensible statement is:

> The recoverable activity associated with the folders SSH, Corel, and Corel Draw Work is concentrated on 24 September 2026. A more precise deletion timeframe cannot be established from the timestamps available in this examination.

Limitations of the Timeline

- No distinct “deletion” timestamp separate from Created/Modified/Accessed was identified.
- Timestamps are very close to the image acquisition time.
- Original full paths (e.g. under a user profile) were not clearly recovered.
- Three target folders produced no recoverable timestamps.

4. **SCREENSHOTS:**

The following screenshots were captured during the examination and are referenced throughout this report:

FIGURE 1: Command used to create working copy of the forensic image and hash verification (SHA-256 and MD5) of the working copy and the original copy.

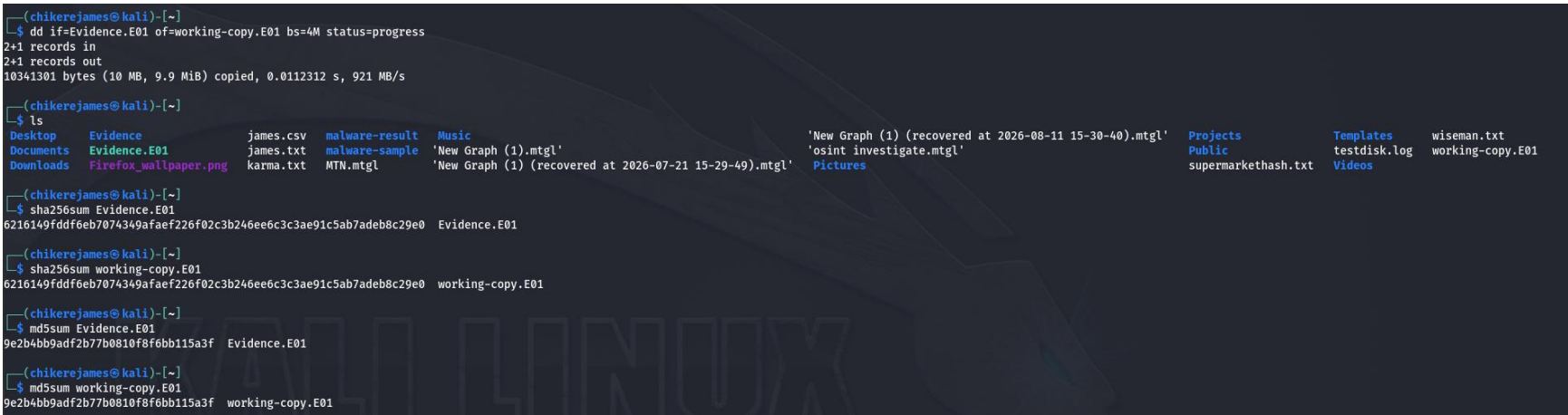


FIGURE 2: FTK Imager-Overview of the loaded forensic image (Evidence.E01) and identification of the file system which is NTFS.

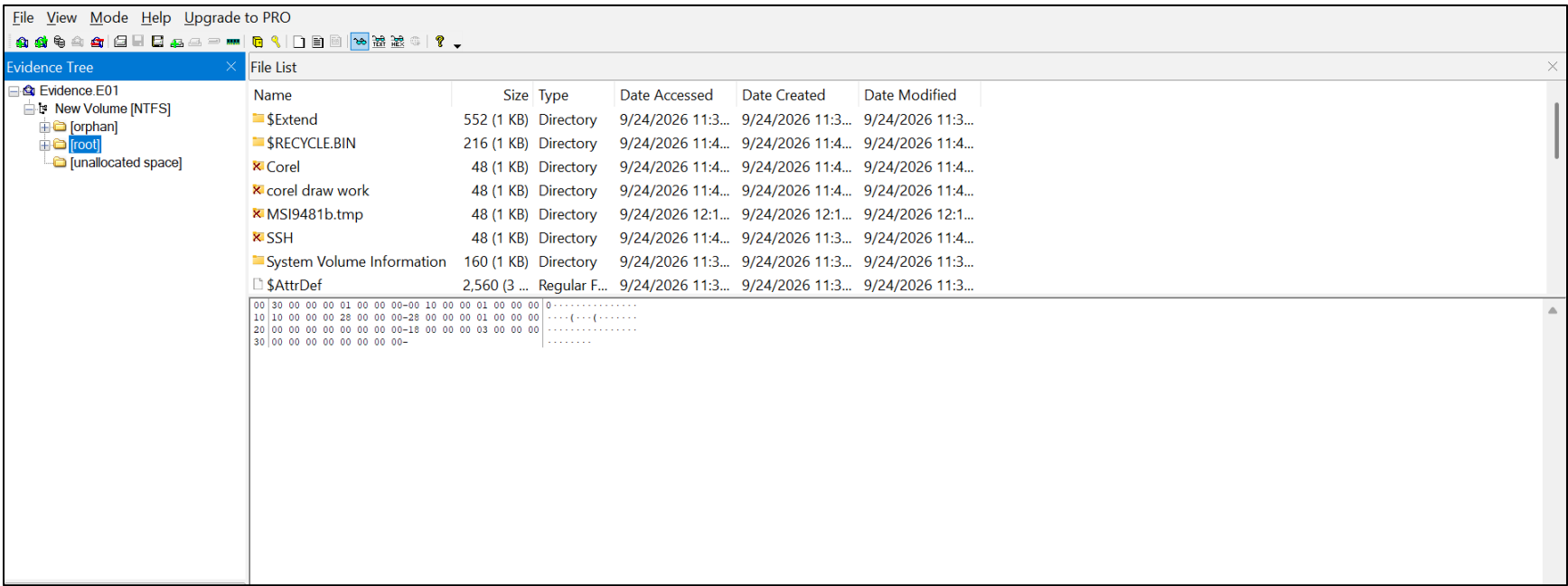


FIGURE 3: FTK Imager-Deleted directory entries identified for the SSH, Corel, and Corel draw work folders.

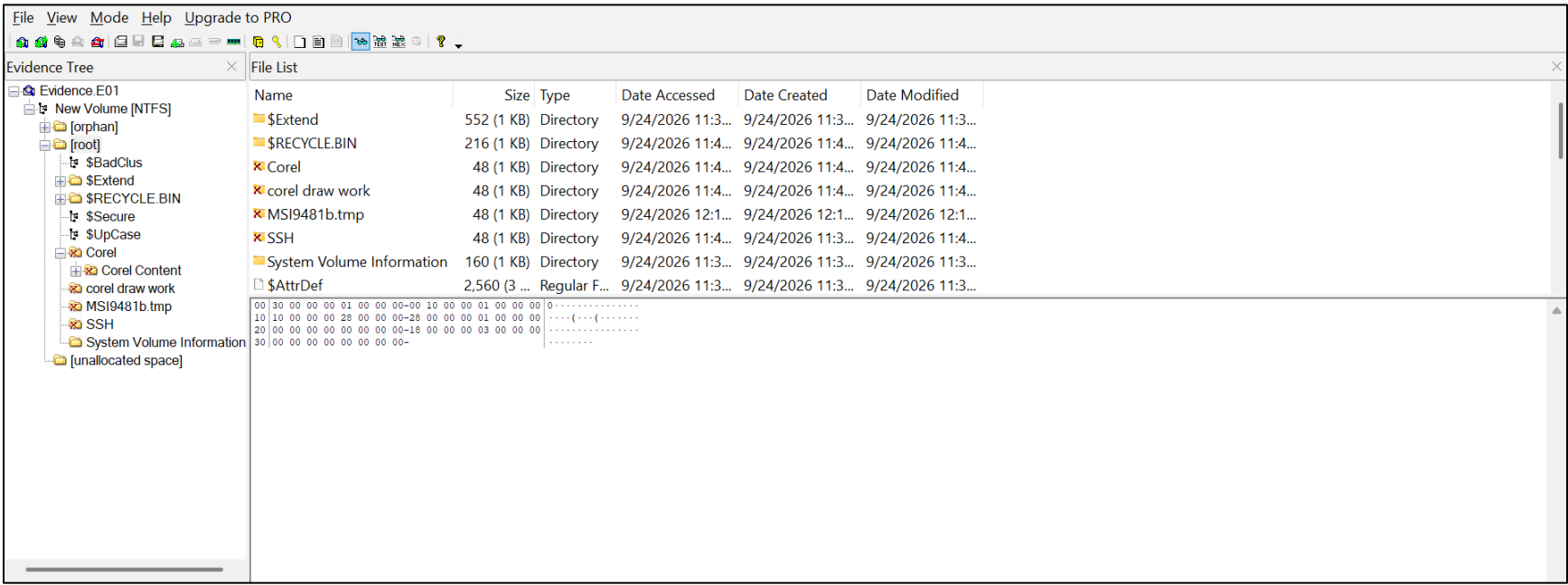


FIGURE 4: Metadata of the forensic image(including size, hash, and acquisition details).

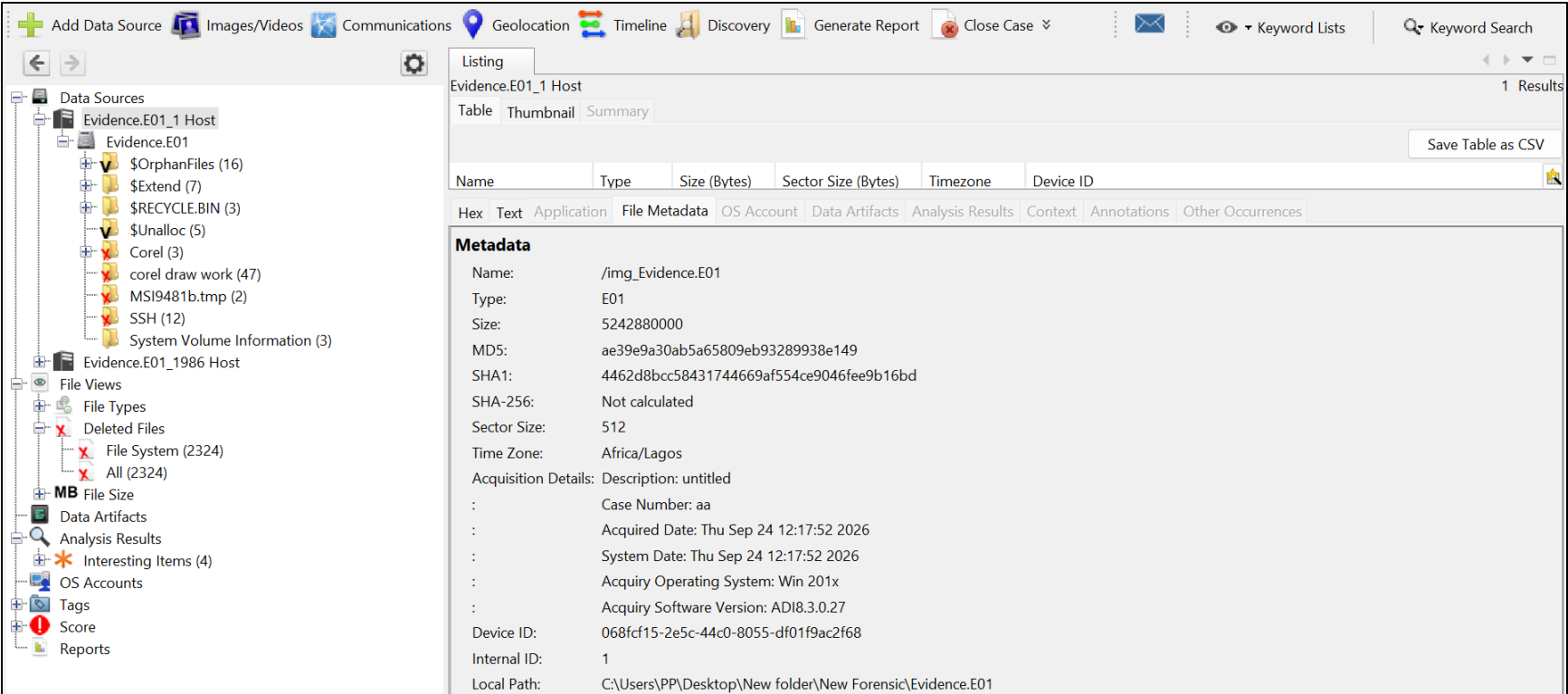


FIGURE 5: Metadata of the identified folder named Corel.

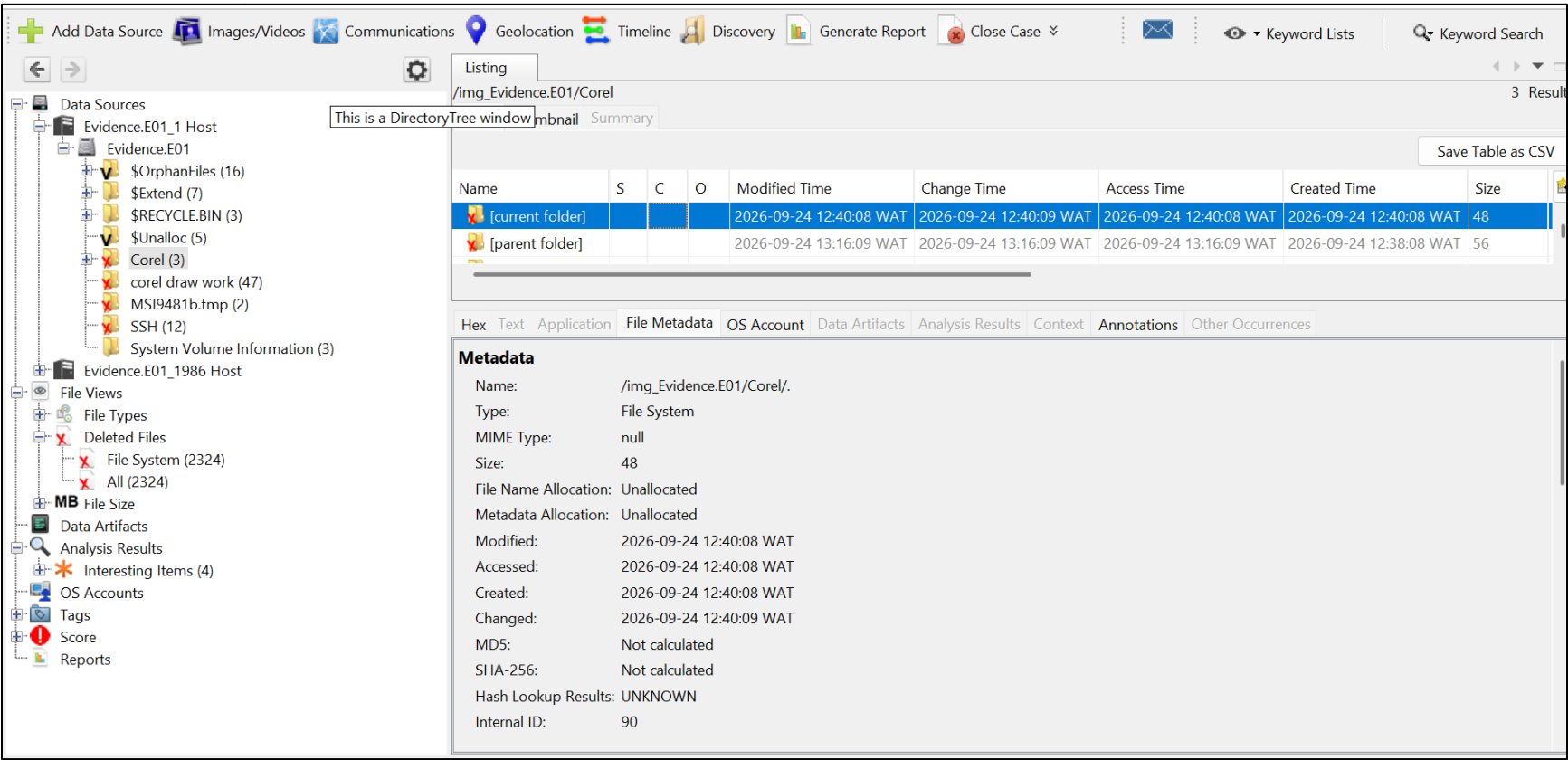


FIGURE 6: Metadata of the identified deleted folder Corel draw work.

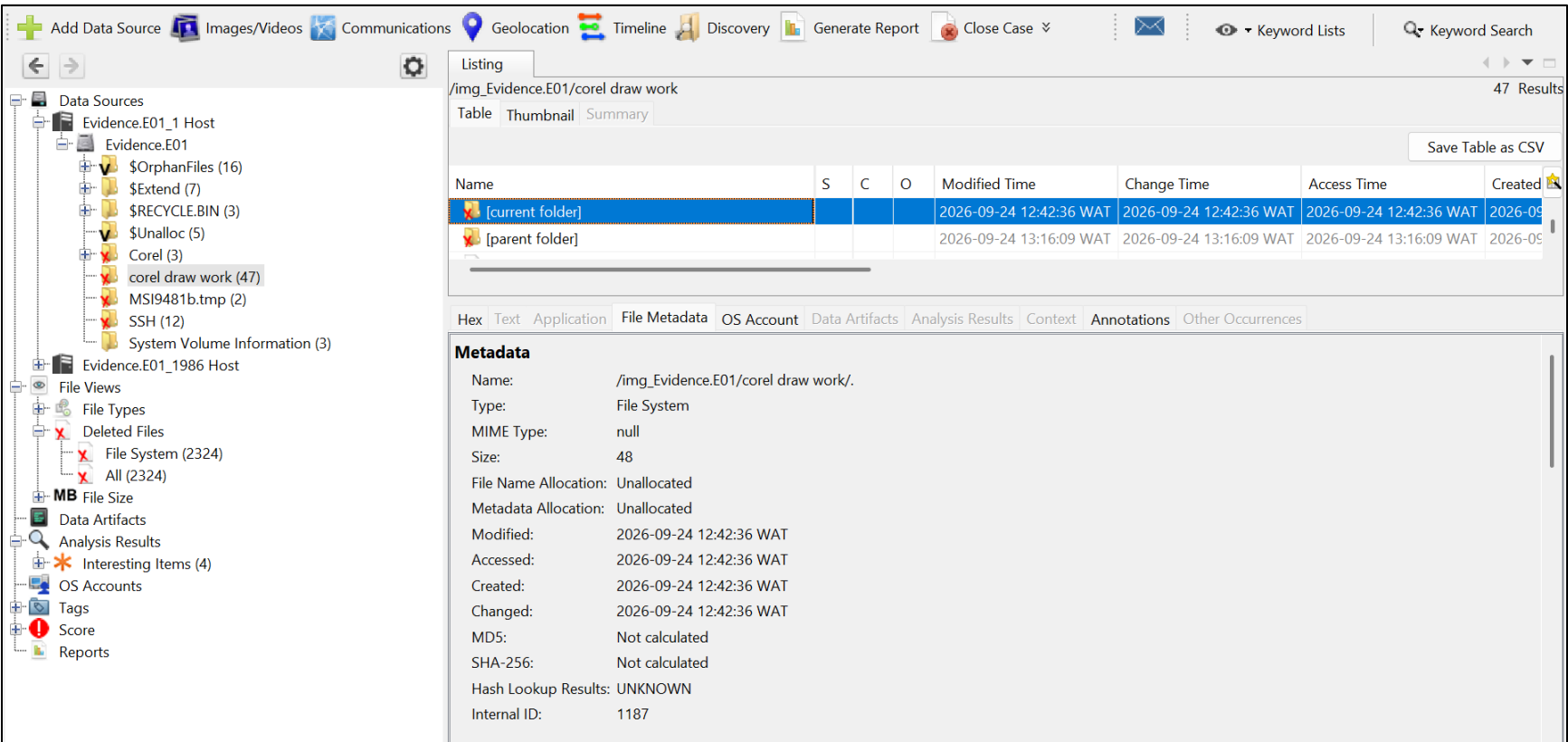


FIGURE 7: Metadata of the identified discovered folder SSH

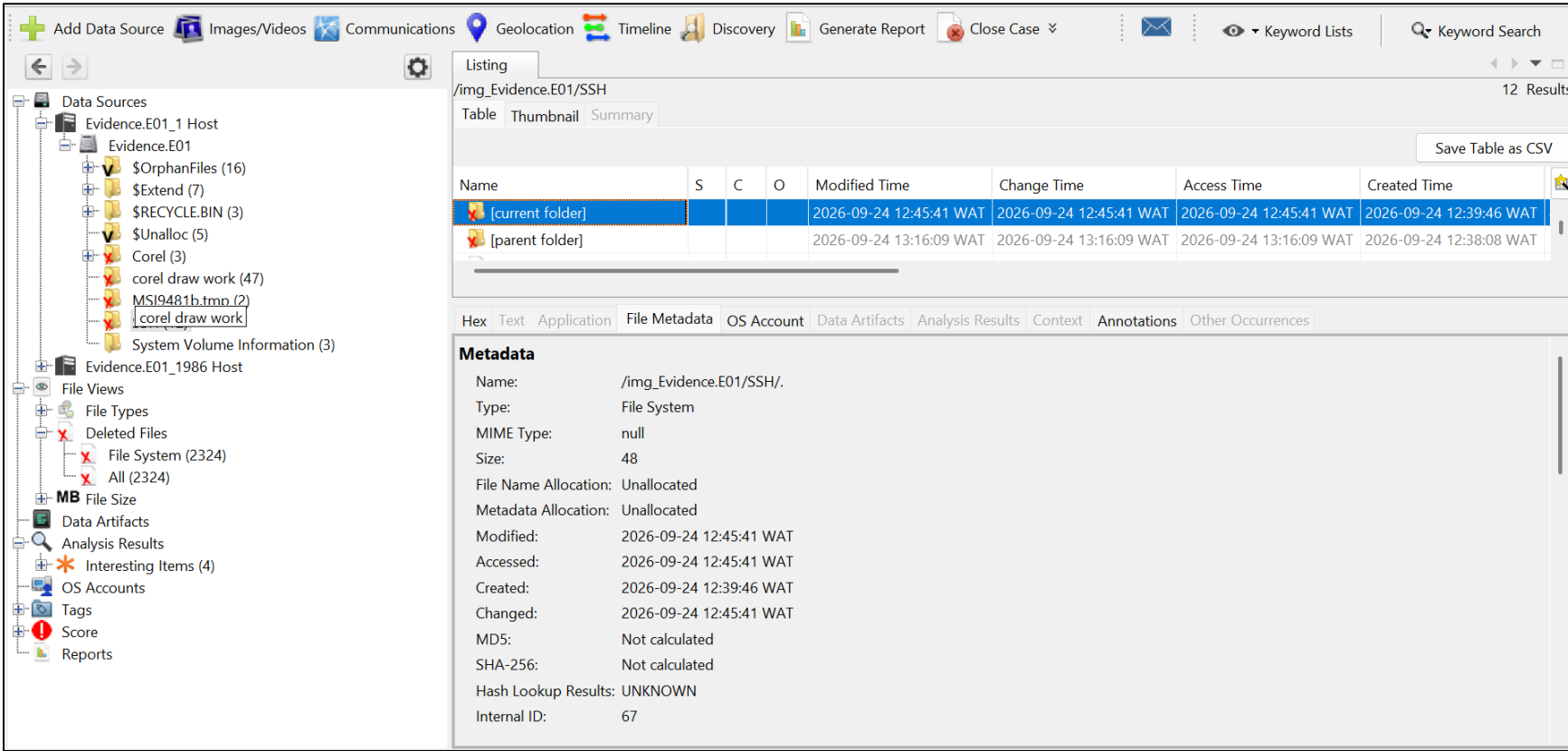


FIGURE 8: Recovered contents of the SSH folder.

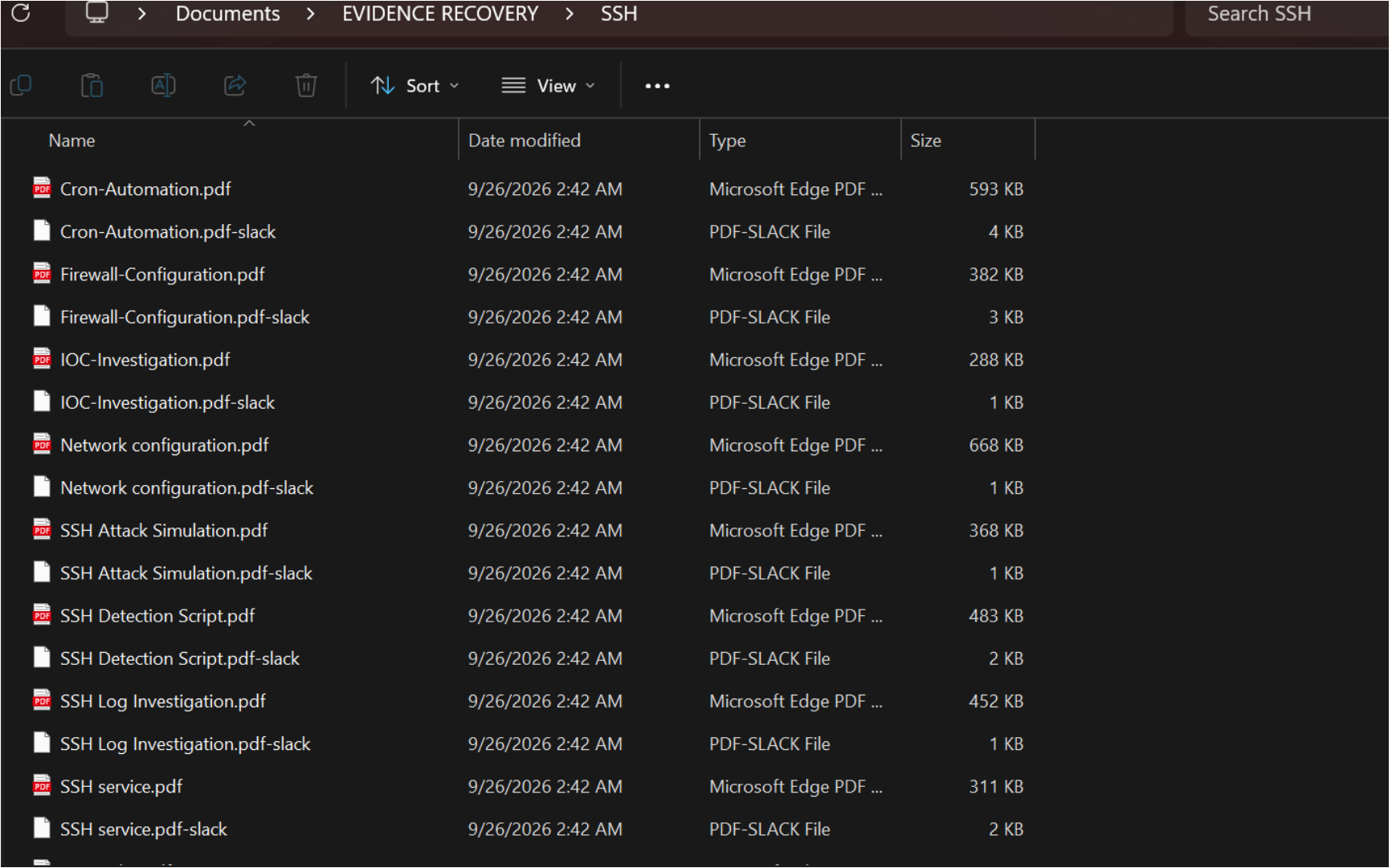


FIGURE 9: Recovered contents of the Corel(corel content) folder.

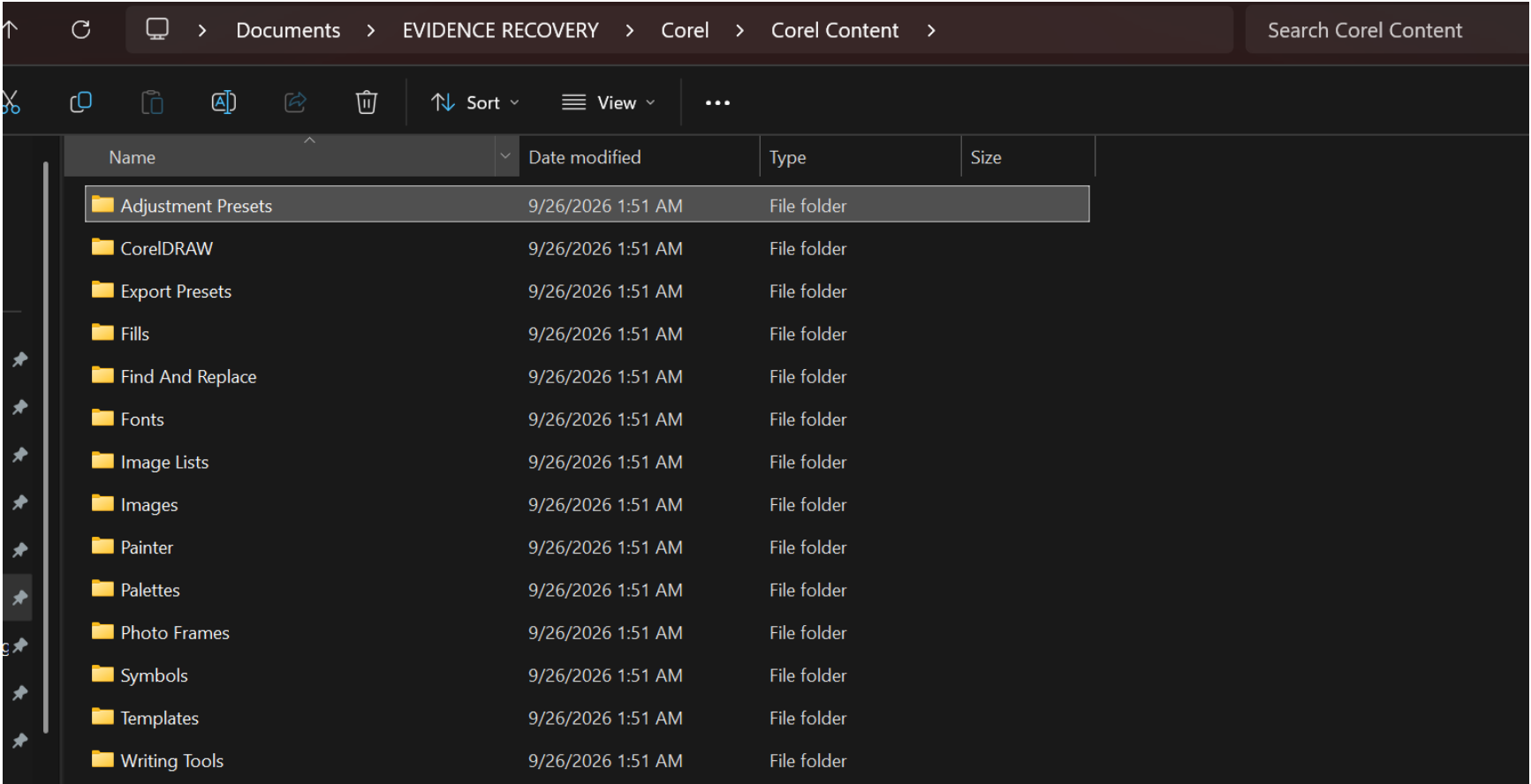


FIGURE 10: Recovered contents of the Corel draw work folder.

Documents > EVIDENCE RECOVERY > corel draw work				
Sort View				
Name	Date modified	Type	Size	
1b11aeba9b6ff7b61d711e838fc7691a.jpg	9/26/2026 1:51 AM	JPG File	62 KB	
1b11aeba9b6ff7b61d711e838fc7691a.jpg-slack	9/26/2026 1:51 AM	JPG-SLACK File	3 KB	
1bd12efa4aff56bc92b683120b21131d.jpg	9/26/2026 1:51 AM	JPG File	20 KB	
1bd12efa4aff56bc92b683120b21131d.jpg-slack	9/26/2026 1:51 AM	JPG-SLACK File	1 KB	
1e6baa03ca277e101c924c6899ca52ef.jpg	9/26/2026 1:51 AM	JPG File	7 KB	
1e6baa03ca277e101c924c6899ca52ef.jpg-slack	9/26/2026 1:51 AM	JPG-SLACK File	2 KB	
3b09eb1f5b3da9e0b10292dc4bd3b529.jpg	9/26/2026 1:51 AM	JPG File	7 KB	
3b09eb1f5b3da9e0b10292dc4bd3b529.jpg-sla...	9/26/2026 1:51 AM	JPG-SLACK File	2 KB	
4.jpg	9/26/2026 1:51 AM	JPG File	25,721 KB	
4.jpg-slack	9/26/2026 1:51 AM	JPG-SLACK File	4 KB	
4a56e8884154f8f138742f360ac5f9b8.jpg	9/26/2026 1:51 AM	JPG File	21 KB	
4a56e8884154f8f138742f360ac5f9b8.jpg-slack	9/26/2026 1:51 AM	JPG-SLACK File	4 KB	
4b4f8ab2317981952d4b9e6088e5abf6.jpg	9/26/2026 1:51 AM	JPG File	80 KB	
4b4f8ab2317981952d4b9e6088e5abf6.jpg-slack	9/26/2026 1:51 AM	JPG-SLACK File	1 KB	
5bb0f73a7b3e0f976acad614a42e5040.jpg	9/26/2026 1:51 AM	JPG File	23 KB	
5bb0f73a7b3e0f976acad614a42e5040.jpg-slack	9/26/2026 1:51 AM	JPG-SLACK File	2 KB	

5. HASH RECORD OF SELECTED RECOVERED FILES:

Only a representative sample of recovered files was hashed due to the large number of items (particularly in Corel Draw Work which contained over 90 files).

FOLD ER	FILENAME	SHA-256 HASH	TOO L
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SSH	Cron-Automation.pdf	b42513949d5d45056f74470d5e146352c7e4fab8aa16e1cb63f1fe47df403050	Autopsy
SSH	Firewall-Configuration.pdf	479c121a430b4f8c988183fb4ae28241186c3d4c8d4bffb1271f70e08cb6075f	Autopsy
Corel	Bright Midtones.cdss	f43e6cf886c4a34ed727240af9fe69bff2cbba12f1f9ee9646963d5bce8b658e	Autopsy
Corel	Dark Midtones.cdss	b2b09a06ed97747871f559b1f8ae399c82856fef4a1af64e5465c2c25363ae3c	
Corel draw work	1b11aeba9b6ff7b61d711e838fc7691a.jpg	ba70a7a532e8e0faf070df731915b154b878b9652410eb45e17cdc2c24f7c0a7	Autopsy
Corel draw work	1bd12efa4aff56bc92b683120b21131d.jpg	3a6333584dc8b35b2d183120e11ff1e81b2ed23914ec2b76a0361a3029bd49cb	Autopsy